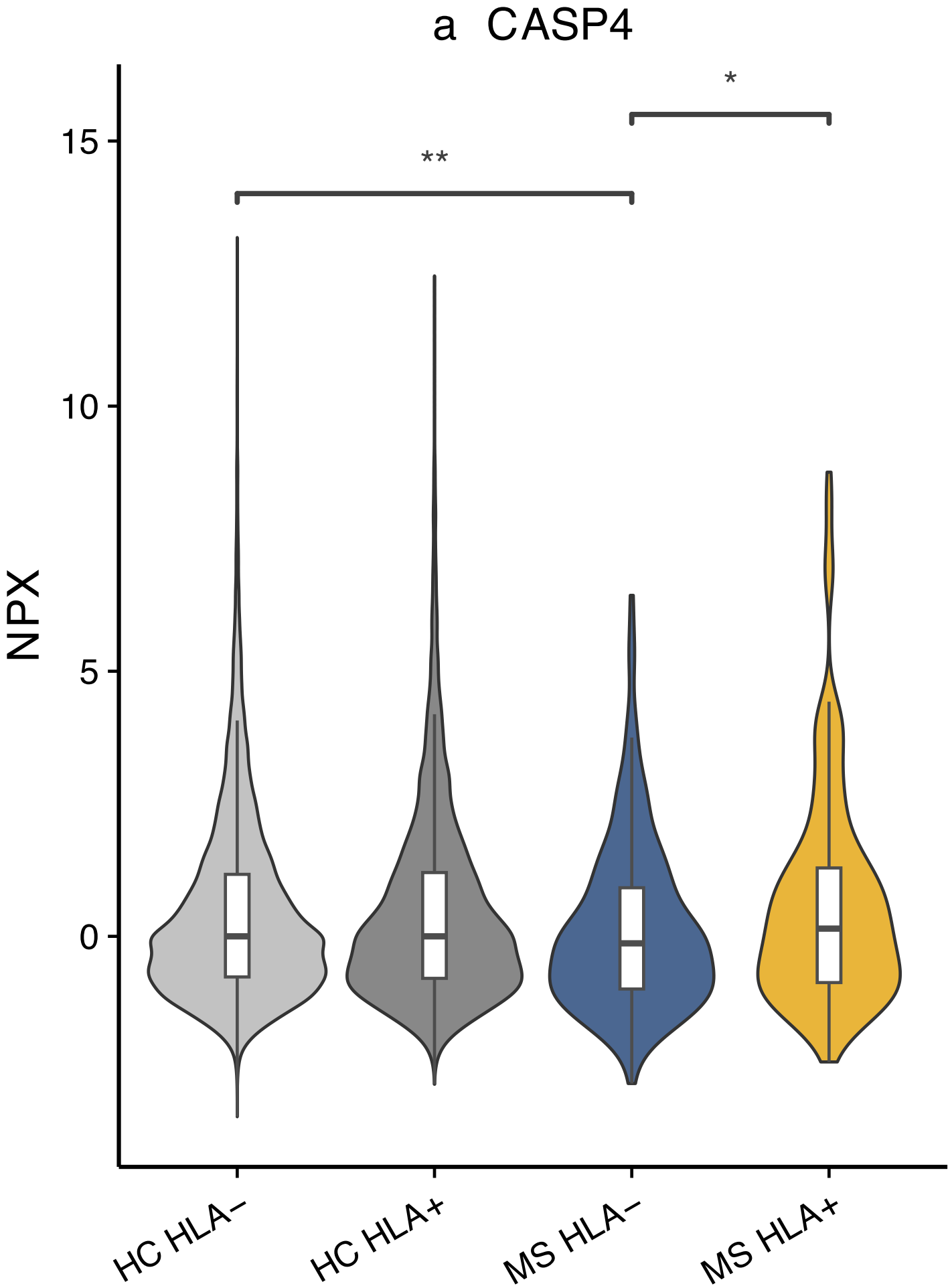
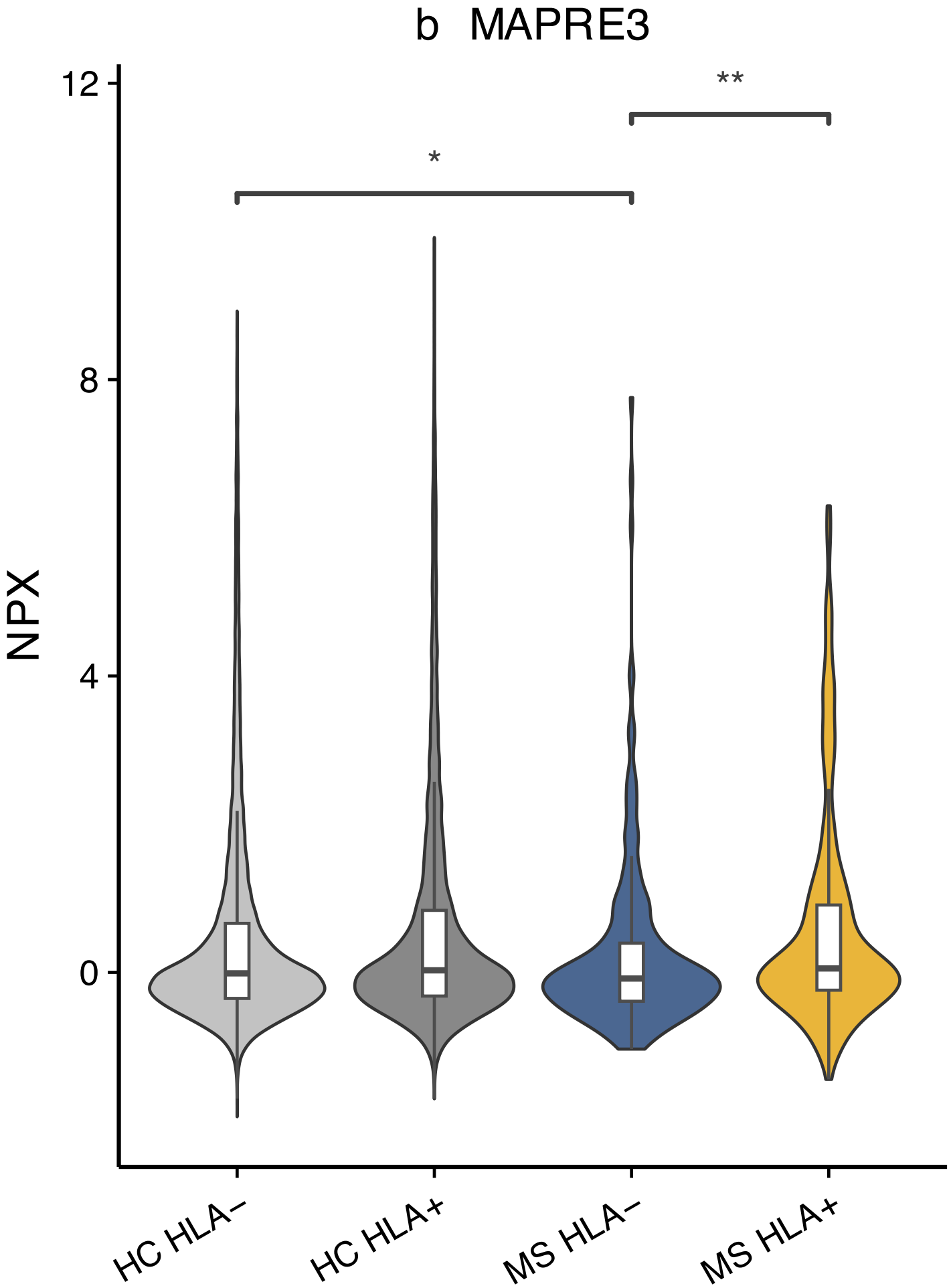
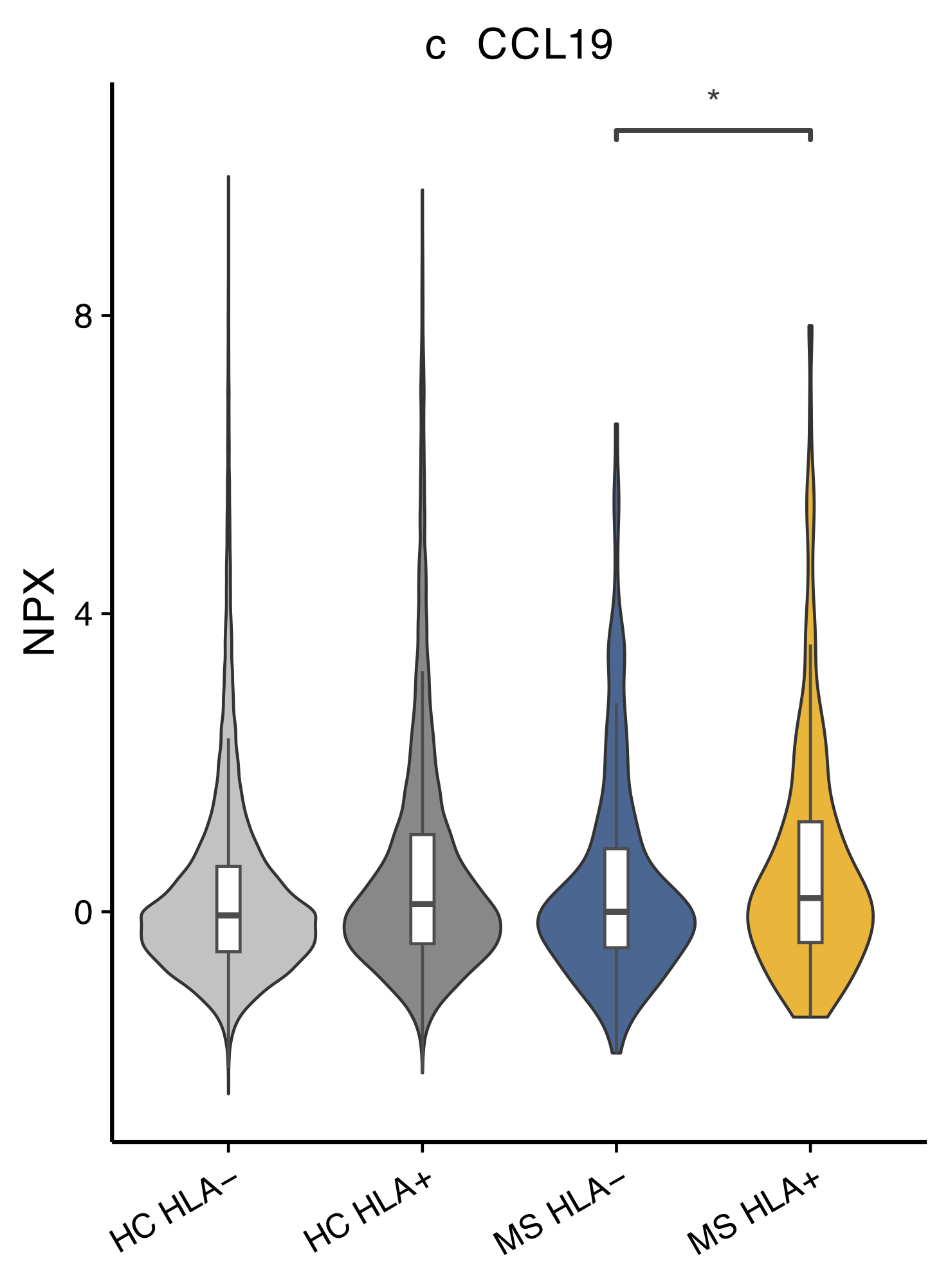
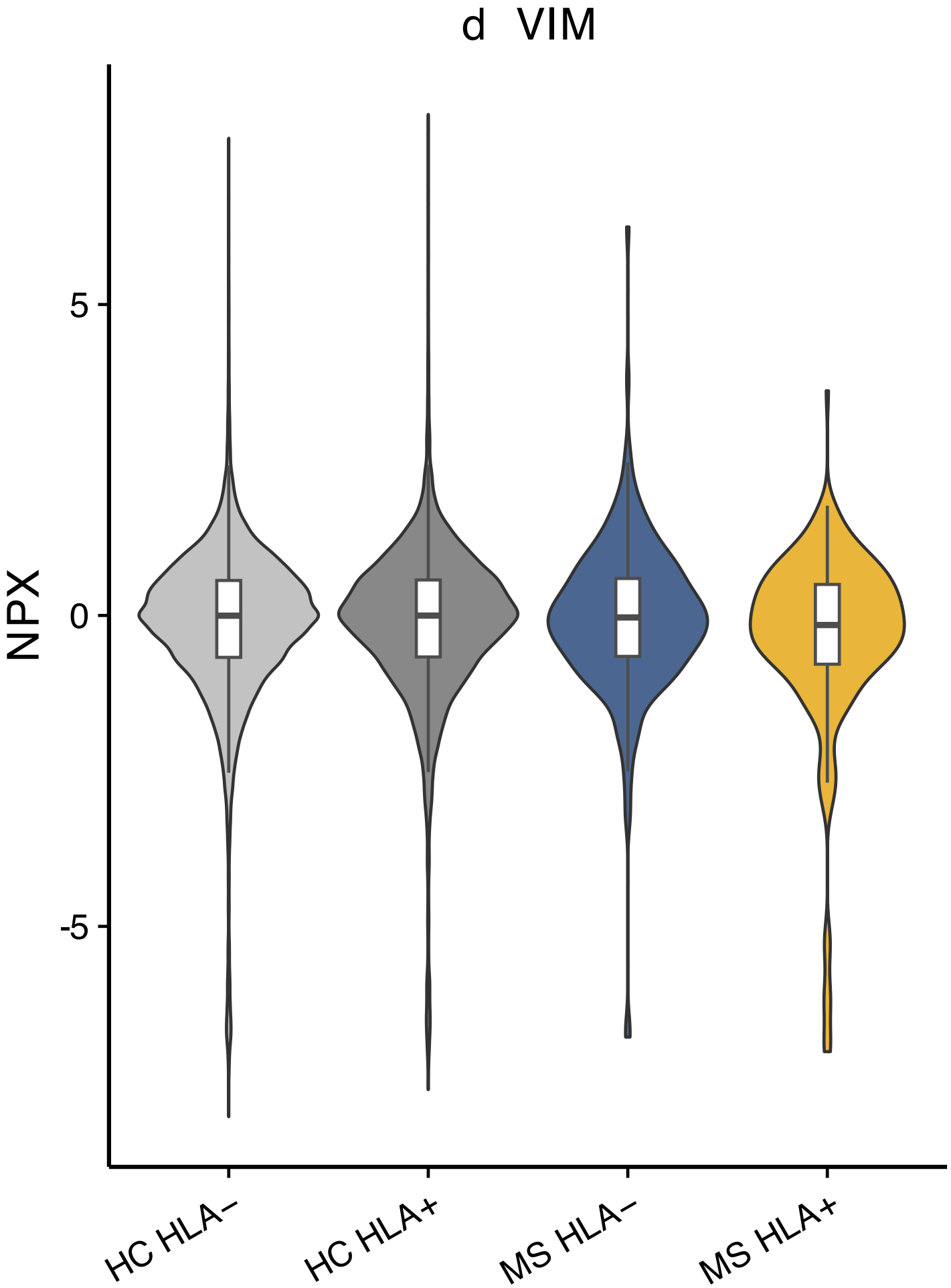
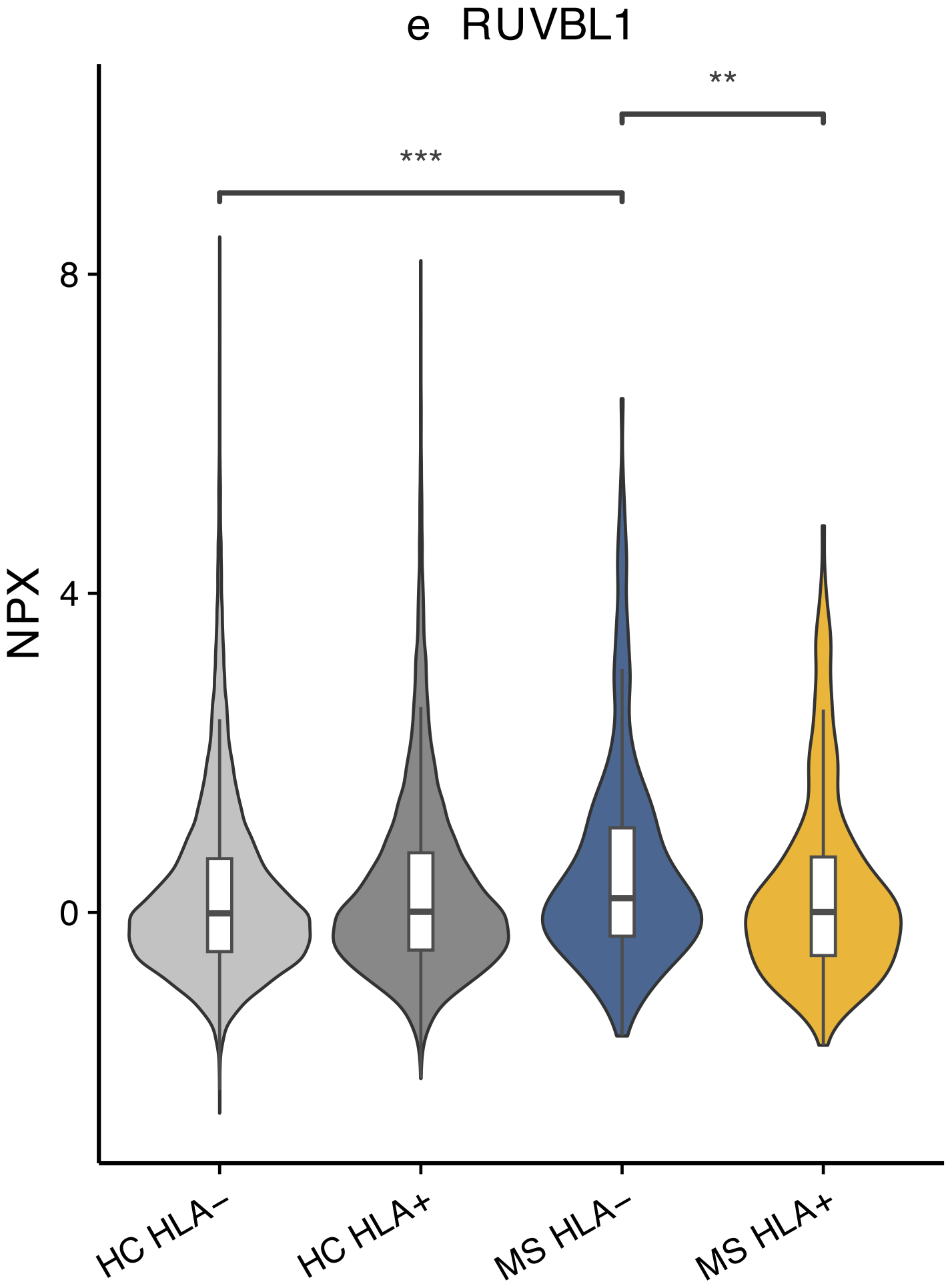
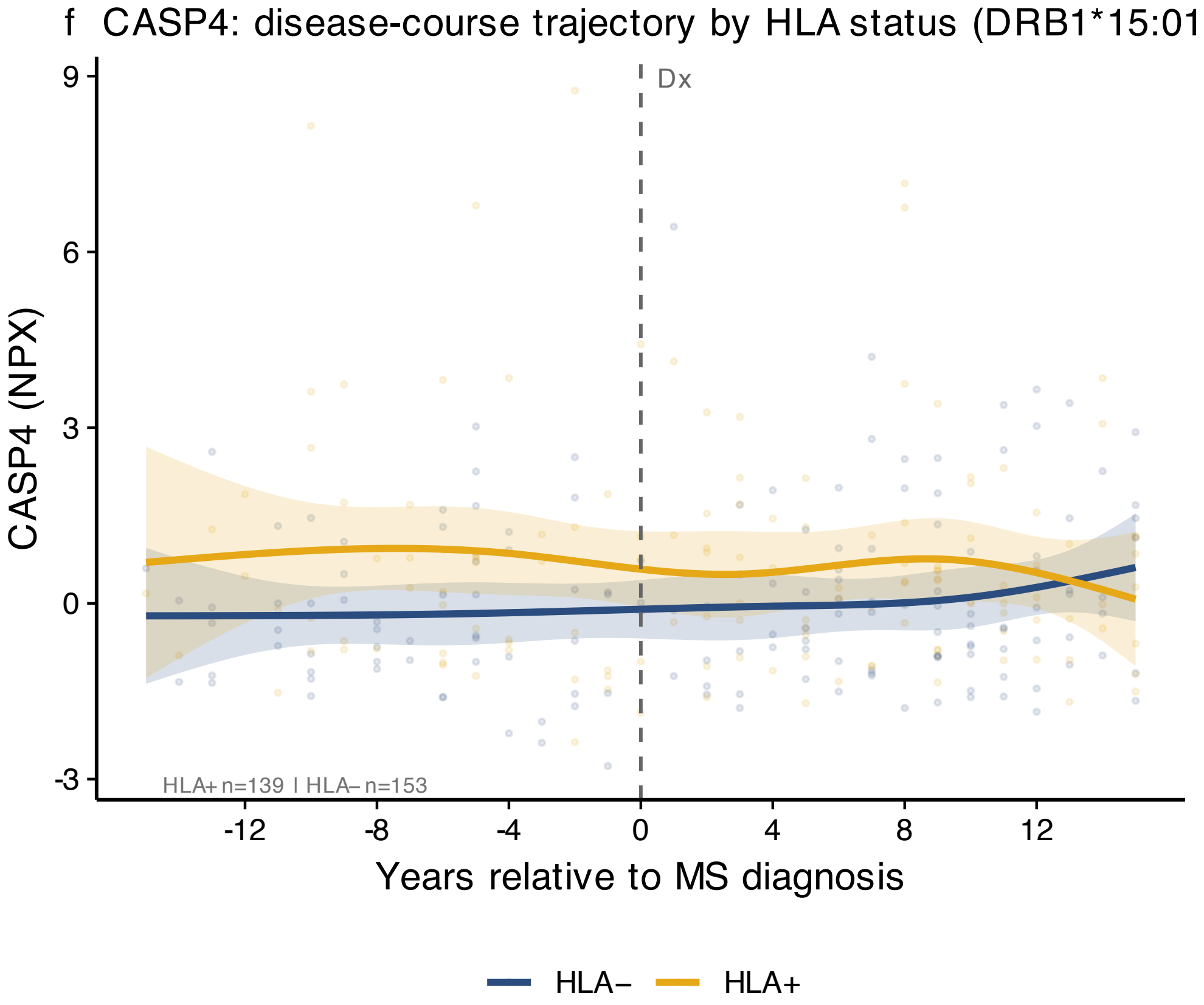

a CASP4
NPX
15
10
5
0
**
*
HC HLA−
HC HLA+
MS HLA−
MS HLA+

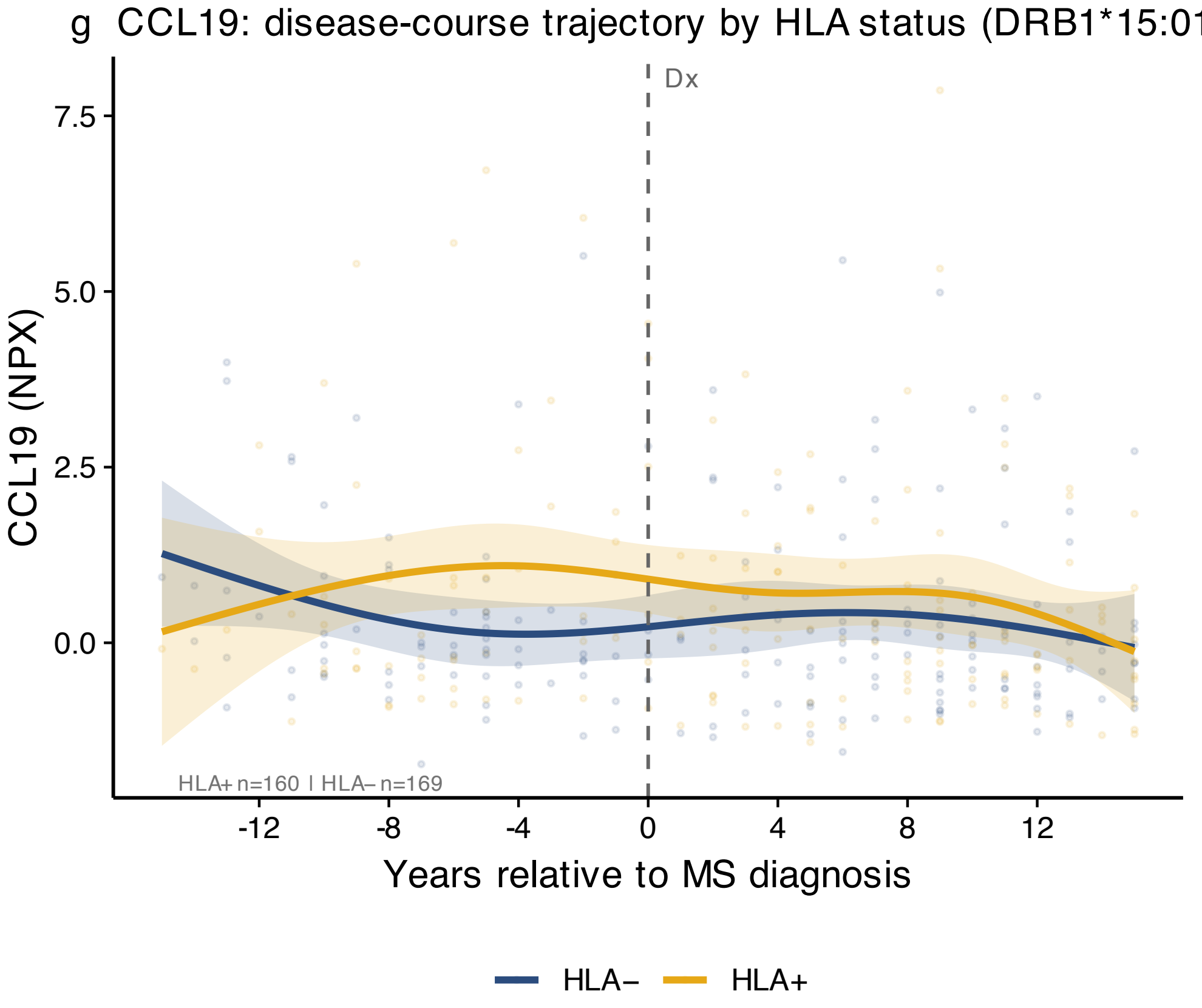

b MAPRE3
NPX
12
8
4
0
*
**
HC HLA−
HC HLA+
MS HLA−
MS HLA+

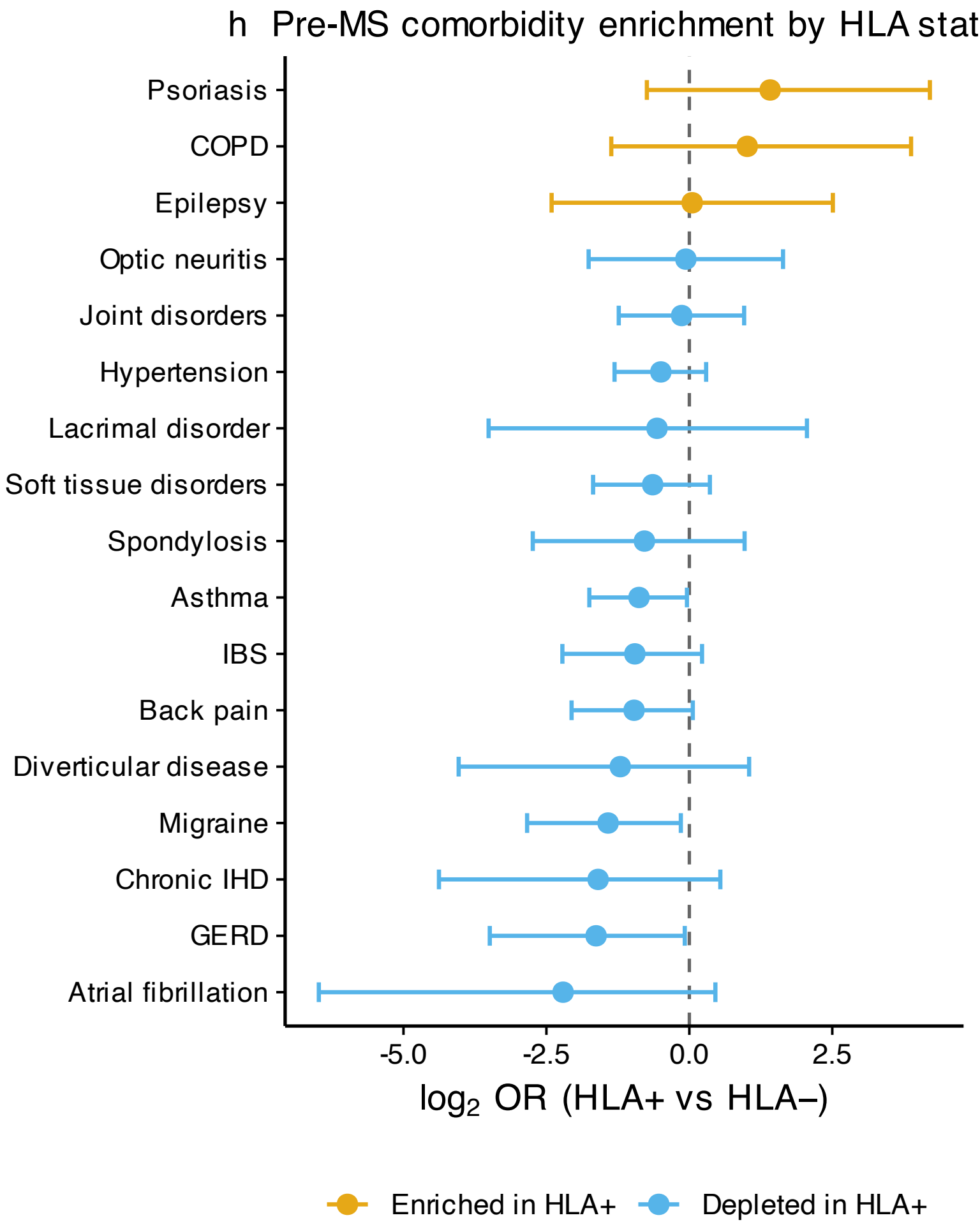

c CCL19
*
8
4
0
NPX
HC HLA−
HC HLA+
MS HLA−
MS HLA+


d VIM
NPX
5
0
-5
HC HLA−
HC HLA+
MS HLA−
MS HLA+


e RUVBL1
**
***
8
4
0
NPX
HC HLA−
HC HLA+
MS HLA−
MS HLA+


f CASP4: disease-course trajectory by HLA status (DRB1*15:01
Dx
CASP4 (NPX)
9
6
3
0
-3
HLA+ n=139 | HLA− n=153
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


g CCL19: disease-course trajectory by HLA status (DRB1*15:01
Dx
7.5
5.0
2.5
0.0
CCL19 (NPX)
HLA+ n=160 | HLA− n=169
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


h Pre-MS comorbidity enrichment by HLA stat
Psoriasis
COPD
Epilepsy
Optic neuritis
Joint disorders
Hypertension
Lacrimal disorder
Soft tissue disorders
Spondylosis
Asthma
IBS
Back pain
Diverticular disease
Migraine
Chronic IHD
GERD
Atrial fibrillation
-5.0
-2.5
0.0
2.5
log2 OR (HLA+ vs HLA–)
Enriched in HLA+
Depleted in HLA+